

Water Body Health Card and Revival Strategies

for Sehore District, Madhya Pradesh — A Remote Sensing and GIS-Based Approach

Course: DSN3099 — Engineering Project in Community Service (UBA Village Adhyan Activity Pool, SEG-1: Water Resource Management)

Problem Statement Reference: PS #1 — "Water body health card and revival strategies"

Scope: District-scale study (Sehore District, Madhya Pradesh)

Faculty Mentor: Dr. Harshlata Vishwakarma, Assistant Professor, School of Computer Science and Engineering, VIT Bhopal

Submission: Review-0 — Problem Identification (Semester V)

1. Team Details

Name	Registration Number	Stream
Sparsh Kapoor	24BAI10017	B.Tech CSE (AI & ML)
Drishti Chopra	24BCY10096	B.Tech CSE (Cyber Security)
Isha Goyal	24BCE10958	B.Tech CSE (Core)
Srijeta Sinha	24BCE11528	B.Tech CSE (Core)
Shreyash Mohanty	24BEC10063	B.Tech ECE
Anshika Jain	24BEC10146	B.Tech ECE

2. Problem Identification

Water bodies across Madhya Pradesh, including in Sehore district, are under increasing pressure from agricultural expansion, siltation, and unregulated catchment-area land use. Sehore district (~6,548 km²) spans five blocks — Sehore, Ashta, Budhni, Ichawar, and Nasrullahganj — and includes at least one block (Ashta) officially categorized as a 'Critical' groundwater zone, with 93.50% groundwater development. Despite this, there is no publicly available, systematic, satellite-derived health assessment of the district's water bodies that would allow district administration, panchayats, and NGOs to prioritize revival interventions such as desilting, catchment treatment, or rainwater harvesting.

This project proposes to build a Water Body Health Card — a structured, per-location assessment combining surface water extent change, vegetation health of the surrounding catchment, and land-use pressure — along with a set of location-specific revival strategies, using open-source satellite data and geospatial tools.

3. Motivation and Community Relevance

- Enables district administration and panchayats to prioritize water body revival with evidence rather than ad hoc selection.
- Supports farmers by clarifying water availability trends relevant to irrigation planning.
- Provides a reusable, low-cost methodology (built entirely on free satellite data and open-source tools) that can be extended to other districts.

- Aligns with the National Water Policy, Jal Jeevan Mission, and the National Geospatial Policy 2022's emphasis on open-source geospatial analysis.

4. Objectives

- Assess water body health in Sehore district using multi-temporal remote sensing data (2017-2024).
- Detect and quantify change in surface water extent, vegetation health, and land use over the study period.
- Identify and rank priority zones for water body revival based on a defined health-scoring methodology.
- Develop an automated, reproducible health-card generation tool rather than a one-time manual report.
- Propose actionable, location-specific revival strategies aligned with national water policy frameworks.
- Validate at least one priority water body through a physical field visit and ground documentation.

5. Proposed Methodology

5.1 Data Sources

Data Type	Source	Resolution	Years
Satellite Imagery	Sentinel-2 (COPERNICUS/S2_SR)	10 m	2017, 2024
Land Use / Land Cover	MODIS MCD12Q1	500 m	2017, 2020
Boundary Data	Survey of India district boundary	-	2023
Vegetation Index (NDVI)	Sentinel-2 derived	10 m	2017, 2024
Water Index (NDWI)	Sentinel-2 derived	10 m	2017, 2024

5.2 Tools and Technology

- Google Earth Engine (GEE) — cloud-based satellite data processing (JavaScript API)
- QGIS — desktop GIS for cartographic output and layout
- Python (Flask, Pandas, ReportLab) — automated health-card and report generation
- All tools used are free and open-source, in line with the project's low-cost, reproducible design goal.

5.3 Analytical Approach

- Water body detection via Normalized Difference Water Index (NDWI) from Sentinel-2 imagery, with pixel-level cloud/shadow masking using the Scene Classification Layer (SCL).
- Vegetation health assessment via Normalized Difference Vegetation Index (NDVI) as an indicator of catchment condition.
- Land Use/Land Cover (LULC) classification via MODIS MCD12Q1, aggregated into Cropland, Forest, Grass/Savannas, Wetland, Urban, and Barren categories.
- Change detection (2017 vs. 2024/2020) to identify water-loss hotspots and catchment degradation.
- A composite health-scoring model combining water extent change, catchment NDVI trend, and adjacent land-use pressure to classify locations as Healthy, Stressed, or Critical.
- Automated generation of a health card per priority location, with revival recommendations mapped to the specific pressures identified at that location.

6. Preliminary Work Completed

As of Review-0, the following technical groundwork has already been completed and will be extended, not restarted, during Phase I:

- District-wide NDWI-based water body detection for 2017 and 2024, with pixel-level cloud masking implemented.
- District-wide NDVI-based vegetation health analysis for 2017 and 2024.
- MODIS-based LULC classification and per-category area statistics for 2017 and 2020.
- Change-detection layers (water change, NDVI change, LULC change) generated as both raster (GeoTIFF) and vector (shapefile) outputs, verified for file integrity.
- Preliminary finding: district-wide water extent increased from approximately 29.45 km² (2017) to 35.33 km² (2024), a net +20% change, while mean NDVI declined marginally (-3.8%), indicating a need for location-level (not only district-level) analysis to identify where losses are concentrated despite the net gain.

Phase I work will focus on converting these existing layers into a structured, scored health-card output and validating priority locations through field visits — rather than on repeating the base remote-sensing analysis.

7. Team Roles

Team Member	Primary Responsibility (Phase I)
Sparsh Kapoor	Remote sensing & GIS pipeline (Google Earth Engine, QGIS), water body health-scoring logic, ML-based classification
Drishti Chopra	Data validation and edge-case testing of health-scoring logic; data integrity / access control for report outputs
Isha Goyal	Automated health-card / report generation (Flask + ReportLab), backend development
Srijeta Sinha	Automated health-card / report generation (Flask + ReportLab), backend development
Shreyash Mohanty	Field-observation data collection tool (mobile-friendly form), field visit documentation
Anshika Jain	Field-observation data collection tool (mobile-friendly form), field visit documentation

8. Field Visit Plan

In line with the course requirement for field visits (within 100 km) to identify community requirements, the team will visit 1-2 priority water bodies identified from the water-loss change-detection map as having the most significant decline, to document siltation, encroachment, and local usage patterns that are not visible from satellite data alone. Findings will be recorded using a lightweight field-data-entry form and incorporated into the final health card and revival strategy for that location.

9. Proposed Timeline (Phase I, Semester V)

Timeline	Activity	Deliverable
Weeks 1-2	Finalize problem scope, mentor consultation, literature/data source review	Review-0 submission
Weeks 3-5	Satellite data acquisition and processing (Sentinel-2, MODIS) via Google Earth Engine; NDWI, NDVI, LULC analysis for the district	Raster layers (water, vegetation, land use) for 2017 and 2024
Weeks 6-7	Change detection analysis; vector export and QGIS-based mapping; identification of water-loss hotspots and priority blocks	Change-detection maps; hotspot shapefiles
Weeks 8-9	Design and implementation of water body health-scoring logic (NDWI change, catchment NDVI, adjacent land use)	Working classification model
Weeks 10-11	Field visit to priority water bodies; ground documentation; development of automated health-card report generator	Field visit report; draft health cards
Weeks 12-13	Consolidation, revival strategy mapping per hotspot, Phase I report preparation	Phase I report (minimum 40% completion)

10. Expected Phase I Outcome (Minimum 40% Completion)

- Working health-scoring methodology applied to real district data, producing classified priority zones.
- At least one completed field visit with documentation.
- Draft automated health-card output (PDF or dashboard) for a subset of priority locations.
- Location-specific revival strategy mapping, extending the six revival-strategy categories already identified (desilting, catchment treatment, rainwater harvesting, afforestation, community monitoring, sustainable agriculture) to specific priority zones.
- Phase I written report.